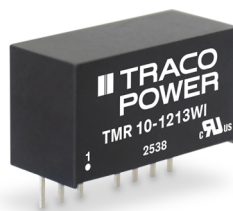


- Ultra compact 10 Watt converter in SIP-8 casing
- High power density of 3,89W/cm<sup>3</sup>
- Wide 4:1 input voltage ranges
- I/O-isolation 1500 VDC
- High efficiency (up to 89%) for low thermal loss
- Operating temperature range -40°C to +70°C (without derating)
- Fully regulated outputs
- Remote On/Off control
- Indefinite short circuit protection
- 3-year product warranty



The TMR 10WI series is a family of isolated 10W DC/DC converter modules with regulated output, featuring wide 4:1 input voltage ranges. The product offers a very high power density of 3.98W/cm<sup>3</sup> in an ultra-compact SIP-8 package occupying only 2.0 cm<sup>2</sup> (0.3 square inch) of board space. An excellent efficiency of up to 89% allows for an extended operating temperature range of -40° to +70°C without derating under natural convection conditions. Further features include remote On/Off control, continuous short circuit protection and an I/O isolation voltage of 1500 VDC. The very compact dimensions of these converters make them an ideal solution for many space critical applications in communication equipment, instrumentation and industrial electronics.

| Models        |                               |          |                  |          |                  |                 |
|---------------|-------------------------------|----------|------------------|----------|------------------|-----------------|
| Order Code    | Input Voltage Range           | Output 1 |                  | Output 2 |                  | Efficiency typ. |
|               |                               | Vnom     | I <sub>max</sub> | Vnom     | I <sub>max</sub> |                 |
| TMR 10-1211WI | 4.5 - 18 VDC<br>(12 VDC nom.) | 5.1 VDC  | 2'000 mA         |          |                  | 89 %            |
| TMR 10-1212WI |                               | 12 VDC   | 833 mA           |          |                  | 89 %            |
| TMR 10-1213WI |                               | 15 VDC   | 666 mA           |          |                  | 89 %            |
| TMR 10-1215WI |                               | 24 VDC   | 416 mA           |          |                  | 89 %            |
| TMR 10-1222WI |                               | +12 VDC  | 416 mA           | -12 VDC  | 416 mA           | 88 %            |
| TMR 10-1223WI |                               | +15 VDC  | 333 mA           | -15 VDC  | 333 mA           | 89 %            |
| TMR 10-2411WI | 9 - 36 VDC<br>(24 VDC nom.)   | 5.1 VDC  | 2'000 mA         |          |                  | 88 %            |
| TMR 10-2412WI |                               | 12 VDC   | 833 mA           |          |                  | 89 %            |
| TMR 10-2413WI |                               | 15 VDC   | 666 mA           |          |                  | 89 %            |
| TMR 10-2415WI |                               | 24 VDC   | 416 mA           |          |                  | 88 %            |
| TMR 10-2422WI |                               | +12 VDC  | 416 mA           | -12 VDC  | 416 mA           | 88 %            |
| TMR 10-2423WI |                               | +15 VDC  | 333 mA           | -15 VDC  | 333 mA           | 88 %            |
| TMR 10-4811WI | 18 - 75 VDC<br>(48 VDC nom.)  | 5.1 VDC  | 2'000 mA         |          |                  | 88 %            |
| TMR 10-4812WI |                               | 12 VDC   | 833 mA           |          |                  | 89 %            |
| TMR 10-4813WI |                               | 15 VDC   | 666 mA           |          |                  | 89 %            |
| TMR 10-4815WI |                               | 24 VDC   | 416 mA           |          |                  | 89 %            |
| TMR 10-4822WI |                               | +12 VDC  | 416 mA           | -12 VDC  | 416 mA           | 89 %            |
| TMR 10-4823WI |                               | +15 VDC  | 333 mA           | -15 VDC  | 333 mA           | 89 %            |

Note - If the input will be switched electromechanically, use an external 68 µF / 100 V / KZE capacitor to reduce voltage transient.

## Input Specifications

|                        |                |                                                                                                                                                                                                  |
|------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input Current          | - At no load   | 12 Vin models: 16 mA typ. / 24 mA max.<br>24 Vin models: 8 mA typ. / 13 mA max.<br>48 Vin models: 5 mA typ. / 10 mA max.                                                                         |
|                        | - At full load | 12 Vin models: 936 mA typ. / 967 mA max.<br>24 Vin models: 473 mA typ. / 494 mA max.<br>48 Vin models: 234 mA typ. / 247 mA max.                                                                 |
| Surge Voltage          |                | 12 Vin models: 25 VDC max. (1 s max.)<br>24 Vin models: 50 VDC max. (1 s max.)<br>48 Vin models: 100 VDC max. (1 s max.)                                                                         |
| Start-up Voltage       |                | 12 Vin models: 4.1 VDC - 4.5 VDC<br>24 Vin models: 8 VDC - 9 VDC<br>48 Vin models: 17 VDC - 18 VDC                                                                                               |
| Under Voltage Lockout  |                | 12 Vin models: 3.7 - 4.3 VDC max.<br>24 Vin models: 7.5 - 8.5 VDC max.<br>48 Vin models: 15.5 - 16.5 VDC max.                                                                                    |
| Recommended Input Fuse |                | 12 Vin models: 5'000 mA (slow blow)<br>24 Vin models: 2'500 mA (slow blow)<br>48 Vin models: 1'500 mA (slow blow)<br>(The need of an external fuse has to be assessed in the final application.) |
| Input Filter           |                | Internal Capacitor                                                                                                                                                                               |

## Output Specifications

|                                        |                                            |                                                                                                                                                                           |
|----------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Voltage Set Accuracy                   |                                            | ±1% max.                                                                                                                                                                  |
| Regulation                             | - Input Variation (Vmin - Vmax)            | single output models: 0.8% max.<br>dual output models: 0.8% max.                                                                                                          |
|                                        | - Load Variation (0 - 100%)                | single output models: 1% max.<br>dual output models: 1% max. (Output 1)<br>1% max. (Output 2)                                                                             |
|                                        | - Voltage Balance (symmetrical load)       | dual output models: 2% max.                                                                                                                                               |
|                                        | - Cross Regulation (25% / 100% asym. load) | dual output models: 5% max.                                                                                                                                               |
| Ripple and Noise<br>(20 MHz Bandwidth) | - single output                            | 5.1 Vout models: 100 mVp-p max. (w/ 1µF)<br>12 Vout models: 140 mVp-p max. (w/ 1µF)<br>15 Vout models: 140 mVp-p max. (w/ 1µF)<br>24 Vout models: 140 mVp-p max. (w/ 1µF) |
|                                        | - dual output                              | 12 / -12 Vout models: 140 / 140 mVp-p max. (w/ 1µF)<br>15 / -15 Vout models: 140 / 140 mVp-p max. (w/ 1µF)                                                                |
|                                        | - single output                            | 5.1 Vout models: 80 mVp-p typ. (w/ 1µF)<br>12 Vout models: 100 mVp-p typ. (w/ 1µF)<br>15 Vout models: 100 mVp-p typ. (w/ 1µF)<br>24 Vout models: 100 mVp-p typ. (w/ 1µF)  |
|                                        | - dual output                              | 12 / -12 Vout models: 100 / 100 mVp-p typ. (w/ 1µF)<br>15 / -15 Vout models: 100 / 100 mVp-p typ. (w/ 1µF)                                                                |
|                                        | - single output                            | 5.1 Vout models: 3'900 µF max.<br>12 Vout models: 680 µF max.<br>15 Vout models: 470 µF max.<br>24 Vout models: 180 µF max.                                               |
|                                        | - dual output                              | 12 / -12 Vout models: 390 / 390 µF max.<br>15 / -15 Vout models: 220 / 220 µF max.                                                                                        |
| Capacitive Load                        |                                            |                                                                                                                                                                           |
| Minimum Load                           |                                            | Not required                                                                                                                                                              |
| Temperature Coefficient                |                                            | ±0.02 %/K max.                                                                                                                                                            |
| Hold-up Time                           |                                            | 1 ms min. (12 Vin models)<br>1.2 ms min. (24 Vin models)<br>5.5 ms min. (48 Vin models)                                                                                   |

All specifications valid at nominal voltage, resistive full load and +25°C after warm-up time, unless otherwise stated.

|                            |                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| Start-up Time              | 30 ms typ. / 50 ms max.                                                                                           |
| Start-up Overshoot Voltage | 5% max.                                                                                                           |
| Short Circuit Protection   | Continuous, Automatic recovery                                                                                    |
| Overload Protection        | Switch off after 4 s delay, automatic restart                                                                     |
| Output Current Limitation  | 110 - 180% of I <sub>out</sub> max.                                                                               |
| Overvoltage Protection     | 140% typ. of V <sub>out</sub> nom.                                                                                |
| Transient Response         | - Response Deviation<br>5% max. (75% to 100% Load Step)<br>- Response Time<br>500 µs max. (75% to 100% Load Step) |

## Safety Specifications

|                       |                                                          |                                                                                                                                                 |
|-----------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Standards             | - IT / Multimedia Equipment<br>- Certification Documents | EN 62368-1<br>IEC 62368-1<br>UL 62368-1<br><a href="http://www.tracopower.com/tmr10wi-safety-cert/">www.tracopower.com/tmr10wi-safety-cert/</a> |
| Energy Source         | - Output, acc. to 62368-1                                | ES1                                                                                                                                             |
| Power Source          | - Output, acc. to 62368-1                                | PS1                                                                                                                                             |
| Pollution Degree      |                                                          | PD 3                                                                                                                                            |
| Over Voltage Category |                                                          | Not mains connected                                                                                                                             |

## EMC Specifications

|                     |                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EMI (Emissions)     | - Conducted Emissions<br>- Radiated Emissions                                                                                          | EN 61000-6-4 (Generic Industrial)<br>EN 55032 class A (with external filter)<br>EN 55032 class A (with external filter)<br>External filter proposal: <a href="http://www.tracopower.com/tmr10wi-emc-filter/">www.tracopower.com/tmr10wi-emc-filter/</a>                                                                                                                                                                                                                                                                              |
| EMS (Immunity)      | - Electrostatic Discharge<br>- RF Electromagnetic Field<br>- EFT (Burst) / Surge<br>- Conducted RF Disturbances<br>- PF Magnetic Field | EN 55035 (Multimedia)<br>Air: EN 61000-4-2, ±8 kV, perf. criteria A<br>Contact: EN 61000-4-2, ±6 kV, perf. criteria A<br>EN 61000-4-3, 10 V/m, perf. criteria A<br>EN 61000-4-4, ±2 kV, perf. criteria A<br>EN 61000-4-5, ±2 kV, perf. criteria A<br>External filter proposal: <a href="http://www.tracopower.com/tmr10wi-emc-filter/">www.tracopower.com/tmr10wi-emc-filter/</a><br>Continuous: EN 61000-4-6, 10 Vrms, perf. criteria A<br>1 s: EN 61000-4-8, 100 A/m, perf. criteria A<br>EN 61000-4-8, 1000 A/m, perf. criteria A |
| EMC / Environmental | - Certification Documents                                                                                                              | <a href="http://www.tracopower.com/tmr10wi-emc-cert/">www.tracopower.com/tmr10wi-emc-cert/</a>                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## General Specifications

|                           |                                                                                                      |                                                                                                                                               |
|---------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Relative Humidity         |                                                                                                      | 95% max. (non condensing)                                                                                                                     |
| Temperature Ranges        | - Operating Temperature<br>- Approved Ambient Temp.<br>- Case Temperature<br>- Storage Temperature   | -40°C to +85°C<br>62368-1: +65°C max. (5.1 Vout models)<br>62368-1: +70°C max. (other models)<br>+105°C max.<br>-50°C to +125°C               |
| Power Derating            | - High Temperature                                                                                   | Depending on model<br>See application note: <a href="http://www.tracopower.com/tmr10wi-cc/">www.tracopower.com/tmr10wi-cc/</a>                |
| Cooling System            |                                                                                                      | Natural convection (20 LFM)                                                                                                                   |
| Remote Control            | - Voltage Controlled Remote (passive = on)<br>- Off Idle Input Current<br>- Remote Pin Input Current | On: 3.5 to 12 VDC or open circuit<br>Off: 0 to 1.2 VDC or short circuit<br>Refers to 'Remote' and '-Vin' Pin<br>0.5 mA max.<br>-0.5 to 0.5 mA |
| Altitude During Operation |                                                                                                      | 6'000 m max.                                                                                                                                  |
| Regulator Topology        |                                                                                                      | Flyback Converter                                                                                                                             |
| Switching Frequency       |                                                                                                      | 390 - 450 kHz (PWM)<br>420 kHz typ. (PWM)                                                                                                     |

All specifications valid at nominal voltage, resistive full load and +25°C after warm-up time, unless otherwise stated.

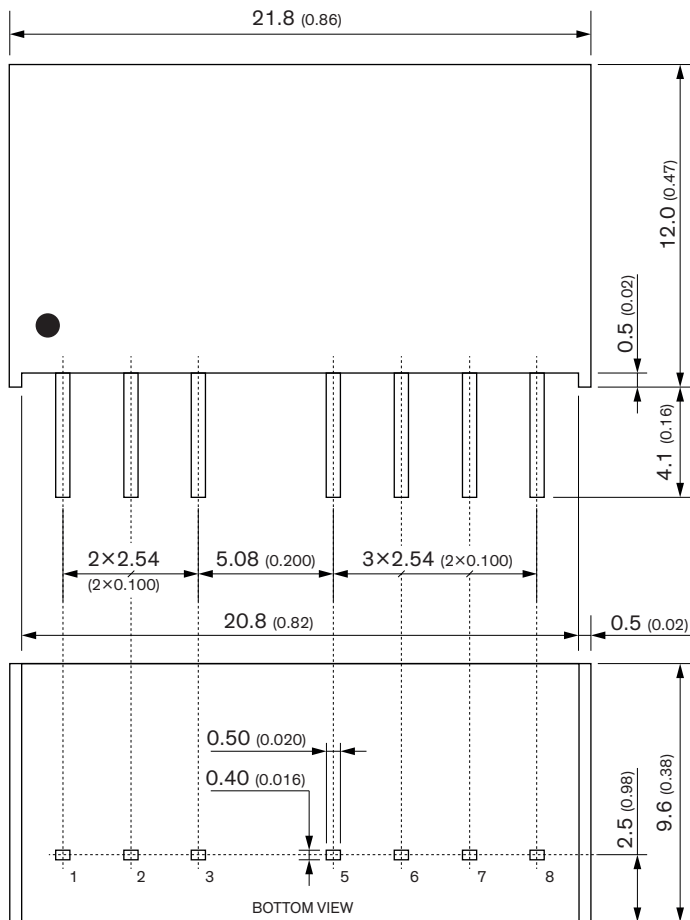
|                            |                                 |                                                                                                                                                                                                                                                                |
|----------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Insulation System          |                                 | Functional Insulation                                                                                                                                                                                                                                          |
| Isolation Test Voltage     | - Input to Output, 60 s         | 1'500 VDC                                                                                                                                                                                                                                                      |
|                            | - Input to Output, 1 s          | 1'800 VDC                                                                                                                                                                                                                                                      |
|                            | - Input to Case, 60 s           | 1'500 VDC                                                                                                                                                                                                                                                      |
|                            | - Output to Case, 60 s          | 1'500 VDC                                                                                                                                                                                                                                                      |
| Creepage                   | - Input to Output               | 0.7 mm min.                                                                                                                                                                                                                                                    |
| Clearance                  | - Input to Output               | 1.5 mm min.                                                                                                                                                                                                                                                    |
| Isolation Resistance       | - Input to Output, 500 VDC      | 1'000 MΩ min.                                                                                                                                                                                                                                                  |
| Isolation Capacitance      | - Input to Output, 100 kHz, 1 V | 1'000 pF typ.                                                                                                                                                                                                                                                  |
|                            |                                 | 1'500 pF max.                                                                                                                                                                                                                                                  |
| Distance Through Isolation |                                 | 1.5 mm                                                                                                                                                                                                                                                         |
| Reliability                | - Calculated MTBF               | 1'590'000 h (MIL-HDBK-217F, ground benign)                                                                                                                                                                                                                     |
| Washing Process            |                                 | According to Cleaning Guideline<br><a href="http://www.tracopower.com/info/cleaning.pdf">www.tracopower.com/info/cleaning.pdf</a>                                                                                                                              |
| Environment                | - Vibration                     | IPC-9592B<br>2.4 g, 3 axis, random waveform, 30 min                                                                                                                                                                                                            |
|                            | - Mechanical Shock              | IPC-9592B<br>30 g, 3 axis, half sine, 11 ms                                                                                                                                                                                                                    |
|                            | - Thermal Shock                 | IPC-9592B<br>-40 to +125°C, 100 cycles, 30 min each                                                                                                                                                                                                            |
|                            |                                 |                                                                                                                                                                                                                                                                |
| Housing Material           |                                 | Plastic (UL 94 V-0 rated)                                                                                                                                                                                                                                      |
| Potting Material           |                                 | Silicone (UL 94 V-0 rated)                                                                                                                                                                                                                                     |
| Pin Material               |                                 | Phosphor Bronze (C5191)                                                                                                                                                                                                                                        |
| Pin Foundation Plating     |                                 | Nickel (1 - 3 μm)                                                                                                                                                                                                                                              |
| Pin Surface Plating        |                                 | Tin (3 - 5 μm), matte                                                                                                                                                                                                                                          |
| Housing Type               |                                 | Plastic Case                                                                                                                                                                                                                                                   |
| Mounting Type              |                                 | PCB Mount                                                                                                                                                                                                                                                      |
| Connection Type            |                                 | THD (Through-Hole Device)                                                                                                                                                                                                                                      |
| Footprint Type             |                                 | SIP8                                                                                                                                                                                                                                                           |
| Soldering Profile          |                                 | Lead-Free Reflow Soldering (acc. J-STD-020E)<br>245°C max. (Tc)<br>10 s max. (tp, at Tp - 5°C)                                                                                                                                                                 |
|                            |                                 | See application note: <a href="http://www.tracopower.com/info/reflow-soldering.pdf">www.tracopower.com/info/reflow-soldering.pdf</a>                                                                                                                           |
| Weight                     |                                 | 7 g                                                                                                                                                                                                                                                            |
| Thermal Impedance          | - Case to Ambient               | 26.39 K/W typ.                                                                                                                                                                                                                                                 |
| Environmental Compliance   | - REACH Declaration             | <a href="http://www.tracopower.com/info/reach-declaration.pdf">www.tracopower.com/info/reach-declaration.pdf</a><br>REACH SVHC list compliant<br>REACH Annex XVII compliant                                                                                    |
|                            | - RoHS Declaration              | <a href="http://www.tracopower.com/info/rohs-declaration.pdf">www.tracopower.com/info/rohs-declaration.pdf</a><br>Exemptions: 7(a)<br>(RoHS exemptions refer to the component concentration only, not to the overall concentration in the product (O5A rule).) |
|                            | - SCIP Reference Number         | be5cadb1-00ed-482a-9818-2ba04826a37f                                                                                                                                                                                                                           |
|                            |                                 |                                                                                                                                                                                                                                                                |

## Additional Information

|                                   |                                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------|
| <b>Supporting Documents</b>       | <a href="http://www.tracopower.com/overview/tmr10wi">www.tracopower.com/overview/tmr10wi</a>   |
| <b>Frequently Asked Questions</b> | <a href="http://www.tracopower.com/glossary-faq">www.tracopower.com/glossary-faq</a>           |
| <b>Glossary</b>                   | <a href="http://www.tracopower.com/info/glossary.pdf">www.tracopower.com/info/glossary.pdf</a> |
| <b>Service Portal</b>             | <a href="http://www.tracoserviceportal.com">www.tracoserviceportal.com</a>                     |

All specifications valid at nominal voltage, resistive full load and +25°C after warm-up time, unless otherwise stated.

## Outline Dimensions

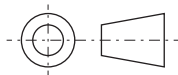


Dimensions in mm (inch)

Tolerances: x.x ±0.5 (x.xx ±0.02)

x.xx  $\pm 0.25$  (x.xxx  $\pm 0.01$ )

Pin tolerances:  $\pm 0.25$  ( $\pm 0.01$ )



| Pinout |               |        |
|--------|---------------|--------|
| Pin    | Single        | Dual   |
| 1      | -Vin          |        |
| 2      | +Vin          |        |
| 3      | Remote On/Off |        |
| 5      | NC            |        |
| 6      | +Vout         |        |
| 7      | -Vout         | Common |
| 8      | NC            | -Vout  |

NC: Not connected

Avoid routing PCB traces under the converter.